BGA 254 UFS 2.1 / 3.1 DIRECT PINOUT & SIGNAL REFERENCE

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BGA 254 PINOUT SPECIFICATION

High-Speed Memory Bus & Power Balls

VCC Balls (K4, L4, M4): 3.3V / 2.8V NAND Flash Memory Array supply

VCCQ Balls (B2, B3): 1.8V Interface Logic Supply

VCCQ2 Balls (C2, D2): 1.2V Low-Voltage Differential Signaling core

DIN_T / DIN_C: Differential input pair (High-speed M-PHY Lane 0)

DOUT_T / DOUT_C: Differential output pair (High-speed M-PHY Lane 0)

RST_N: Active-low hardware reset signal from Application Processor

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UFS LUN PARTITION MAP

Logical Unit Number Layout for Android Boot

LUN 0: User Area (Ext4 / F2FS file system with user data)

LUN 1 (Boot A): Secondary Boot Loader (xbl_a, abl_a, tz_a)

LUN 2 (Boot B): Redundant Boot Loader (xbl_b, abl_b, tz_b)

LUN 3: RPMB (Replay Protected Memory Block) - Cryptographic pairing keys

RPMB Counter: Write counter cannot be reset; mismatch causes Qualcomm 9008 loop

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